RITVIK AGGARWAL

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EDUCATION & HONORS

Indiana University, Luddy School of Informatics, Computing and Engineering *Dec 2026*

Bachelor of Science in Informatics, Minor in Business and Entrepreneurship — GPA: 3.85/4.00

- Dean's List; member of Alpha Lambda Delta and the National Society of Collegiate Scholars for academic standing
- Selected for the Luddy Academic Integrity Council; two-time competitor at the National Swimming Championship

TECHNICAL SKILLS

Python, SQL, Java, LangChain, Mistral AI, Microsoft Graph API, RAG, TensorFlow, PyTorch, AWS, Azure OpenAI, Docker, Git

PROFESSIONAL EXPERIENCE

GitHub | Independent Contributor *August 2026 - Present*

Bloomington, IN

- Debugged 15+ Python training and inference scripts by tracing failures in data loading, tokenization, and batching, then rerunning evaluations
- Added 30+ tests for empty generations, malformed JSON outputs, API timeouts, and missing inputs to catch failures early and improve reliability
- Fixed version conflicts in requirements files and documented setup so contributors could reproduce training and inference runs locally anywhere
- Shipped fixes through Git branches and pull requests with clear commit messages and linked issues, making changes easy for maintainers to review

Snapchat Lens Studio | Independent AR Lens Creator *August 2026 - Present*

Bloomington, IN

- Designed and published 5+ original AR lenses in Lens Studio, each approved through Snap's review process for public release on Snapchat
- Built face-tracking and world-tracking effects with JavaScript scripting, 3D assets, and real-time computer vision features for interactive experiences
- Iterated on lens design using viewer engagement and feedback, reaching 50K+ total views across all published lenses on Snapchat
- Tested lenses on iOS and Android devices and optimized asset size and frame rate to meet Snap's submission guidelines and device requirements

EXL | Analytics and AI Solutions Intern *June 2026 - August 2026*

New York City, NY

- Building a ML pipeline that processes 5K+ customer complaint call transcripts, cleaning and preparing the data for model training and analysis
- Analyzing conversations to identify complaint patterns and behavioral signals that may indicate a customer is considering canceling their Amex card
- Training a predictive model on transcript data to estimate when a customer will cancel their card based on interaction patterns, reaching 82% F1
- Developing logic that turns insights from calls into cancellation-risk predictions, helping identify at-risk customers before they cancel their cards

School of AI | Machine Learning Intern *December 2025 – April 2026*

Bloomington, IN

- Building an end-to-end churn prediction pipeline using tenure, billing, and contract features, improving model F1 score through iterative tuning
- Automating F1 and precision-recall evaluation across 3 ML pipelines, catching model degradation before it reached production
- Handling severe class imbalance in a fraud detection model with class weighting and undersampling, improving detection of rare fraud cases
- Delivering 25 ML projects during the internship, reusing a consistent preprocessing and evaluation pipeline to speed up every new build

Deloitte | Gen AI Intern *May 2025 – August 2025*

Gurugram, India

- Building the backend pipeline that parses uploaded PDF, Word, and Excel files, extracting text for the AI model to answer user questions
- Chunking and embedding extracted document text into a vector database, enabling fast retrieval across a growing document collection
- Building retrieval logic that matches user questions to the closest document chunks, grounding AI answers instead of relying on guesses
- Improving backend logic and frontend responsiveness, boosting overall application performance by 30% for enterprise clients

PROJECTS

AI Trading & Behavior Modeling *Spring 2026*

- Built an LSTM-Transformer ensemble in PyTorch, fusing bidirectional LSTM and positional-encoded Transformer outputs to predict returns
- Backtested signals out-of-sample with configurable transaction costs, tracking Sharpe ratio, drawdown, hit rate, and a confusion matrix
- Built a bidirectional LSTM with attention pooling in PyTorch, classifying user state into 4 categories from 10 behavioral features
- Added an HMM-style transition layer to smooth predictions across a session, then deployed a live FastAPI demo on Render

LEADERSHIP & INVOLVEMENT

Professional Readiness at IU | Co-President *August 2024 – December 2024*

Bloomington, IN

- Led 10+ workshops on business etiquette and professional development for 30+ students each semester, covering interviews and networking
- Recruited guest speakers from 5+ industries for sessions with 50+ participants, and managed fundraising for venues and materials

Middle Eastern Business Association | Co-President *January 2024 – Present*

Bloomington, IN

- Partnered with 5+ campus organizations to co-host inclusive cultural and business events, coordinating logistics and cross-club scheduling
- Led marketing and outreach efforts that grew club membership and attendance through social media campaigns and in-person tabling

INTERESTS

Computer Vision | Swimming | Badminton | Traveling | Vibe Coding | Automation